



Degree project in technology

First cycle 15 credits

Impact of shot length and motion on cinematic tempo

FELIX BROBERG

ANDREAS PANAGIOTIDIS

Impact of Shot Length and Motion on Cinematic Tempo

Felix Broberg
KTH Royale Institute of Technology
Stockholm, Sweden
fbrobe@kth.se

Andreas Panagiotidis
KTH Royale Institute of Technology
Stockholm, Sweden
panagi@kth.se

ABSTRACT

Tempo is an important part of film, it is used by filmmakers to communicate various feelings and settings through visual media. Through research, we found shot length and motion are what primarily makes the tempo of a film. Further it was discovered that over the last 80 years the average shot length has steadily decreased while the action genre has grown over the last 30 years. With these findings the purpose of this study became to examine how the shot length and motion impact the cinematic tempo in action films.

The first part of the study is about trying to estimate the tempo of a film sequence. This was done by finding suitable equations for tempo in films, the equation used needed two parameters to work, the average shot length of the video and its average motion. The motion was computed by extracting motion vectors of the movement from one frame to another, and then calculating the average motion throughout the scene's entirety. The motion value was later combined with the average time of a shot to get the tempo of the scene.

These calculations were carried out on five different action films, two were extreme cases in regard to their average shot length. The films would later be shown to test subjects, who would answer questions about tempo, feelings, preference, et cetera.

The results found from showing the films to test subjects was that shot length was the most important factor in how viewers perceive cinematic tempo. Motion on the contrary did not have as big of an impact. Ideally, we would not have included the extremes regarding shot length when showing the films, as the results would have provided more useful information.

SAMMANFATTNING

Tempo är en viktig del av film, det används av filmskapare för att förmedla olika känslor och tonsättningar via visuell media. Tidigare forskning har visat att klipplängd och rörelse är primärt vad som bidrar till en films tempo. Vi upptäckte även att klipplängden stadigt har minskat de senaste 80 åren medan actiongenren har ökat de senaste 30 åren. Med dessa upptäckter blev studiens syfte att undersöka hur klipplängd och rörelse påverkar filmtempo i actionfilmer.

Första delen av studien gick ut på att estimerat tempot av en filmsekvens. Detta gjordes genom att hitta lämpliga ekvationer för tempo i film, ekvationen som användes behövde två parametrar för

att fungera, genomsnittliga klipplängden samt genomsnittliga rörelsen. Rörelsen beräknades genom att extrahera rörelsevektorer mellan bildramar för att sedan beräkna den genomsnittliga rörelsen genom hela filmsekvensen. Värdet som erhöles användes senare i kombination med den genomsnittliga klipplängden för att få tempot på filmsekvensen.

Dessa beräkningar gjordes på fem olika actionfilmer varav två stycken var extremfall gällande deras genomsnittliga klipplängd. Filmerna visades sedan till försökspersoner som sedan svarade på frågor angående tempo, känslor, preferenser, et cetera.

Resultaten visade att klipplängden var den viktigaste faktorn för hur tittare uppfattar filmtempo. Rörelse hade däremot inte lika stor påverkan. Idealt hade vi inte inkluderat våra två extremfall när filmsekvenserna visades för försökspersonerna då resultaten hade gett mer användbar information.

Keywords

Film; Tempo; ASL; Motion magnitude

1. INTRODUCTION

Filming and editing go hand in hand when creating the tempo of a film. Over the course of time, various editing and filming styles have emerged to contribute to a higher or lower tempo. With time the tempo of movies has increased, this is partially due to shorter shot durations as there has been a linear decline in average shot lengths for about eighty years [4][3]. With such a prominent decline, there is a conspicuous difference in tempo in today's films compared to earlier films. However, no definitive conclusion has been established as to why this linear decline has occurred. It does suggest the significance shot duration might have when creating or watching film, especially regarding tempo.

Over the last thirty years the action genre has steadily grown and stood for 51.3 % of market share in 2021, a big contrast from 18.1% in 1995[2]. The rapid growth of the genre makes it relevant for the study. In addition, action movies are also known for being very fast and intense when conveying the action. Action scenes in different films could therefore be seen as similar in the sense of creating

intensity and capturing viewers' attention. This makes action scenes relevant for the study as they could be comparable in this regard.

1.1 Tempo in Film

Tempo or pace is a fundamental aspect of film created by the movement in the film and what the audience then perceives as either fast or slow. Tempo in film is described as being dictated by three factors, the speed and the rhythm of movement and cuts, the accompanying music, and the overall content [1]. In this study, we will only focus on the first factor, movement and cuts. This is due to the focus of the study being on shorter sequences from films and not full-length films, which makes the tempo contribution from story and content irrelevant. There is also the large focus on editing and visuals, while not taking sound into consideration.

Shorter shot lengths have long been a tool to help contribute to a fast tempo, especially in action movies. It's also a tool to hide imperfections that the audience is not supposed to see, for example stunt doubles, choreography errors etc. Even though fast cuts are a known tool to change how we perceive the film's tempo, many similar action scenes with high tempo have drastically different editing and shot lengths. For example, "*Taken 3*" [11] with its very quick shots, and "*Dunkirk*" [12] with its more drawn-out shots.

Some action movies make use of very fast cuts, while others take longer time with each shot. Faster cutting is a way to achieve a high tempo but other factors such as movement and sound also drive a high tempo and intensity of the scene, which would lead to a better viewing experience for action and gain stronger emotional responses common in the action genre.

1.2 Problem and Hypothesis

The purpose of this study is to investigate how viewers, with today's increased tempo of action movies, perceive films of the same genre with varying editing styles and how it impacts the perceived tempo and viewing experience.

The hypothesis for the study would be that if film sequences have similar tempo, shot length would have the biggest impact on how the viewer perceives the tempo.

For the experiment itself the plan is to show the test subjects different action scenes. All scenes are in high tempo with different shot length. Test subjects will then be asked questions about how they perceived each scene with regards to tempo, what emotions it invoked and which scene they preferred.

2. BACKGROUND

In this section, all theory and related work will be presented to provide a basis for the rest of the study.

2.1 Previous Studies

In a previous study about shot structure in Hollywood, films from three different genres (drama, action, comedy) were compared with regards to structure of shots and shot length. The films were from 1945, 1965, 1985 and 2005, all of which were top grossing films of the year in said category. It is argued that technology has played a part in the decrease of shot length over time, but the viewer's interest also played a part. With shorter shot lengths it is possible

to have more information displayed on screen in the same amount of time, which viewers have gotten used to [18]. This could explain why older movies with longer shot lengths are perceived as less interesting by the average movie-goer.

2.2 Average Shot Length

ASL (average shot length) is a measurement of how long the shots are on average throughout a film or video. This value is attained by adding the length in seconds of all shots together, and then dividing by the number of shots. The term ASL is used by Grubb and Gannon [20] in their article on ASL effect on the viewer's attention and eye movement.

2.3 Editing

When editing and assembling different clips into a longer sequence or film it is called montage [17] and it is mainly carried out by the director and editor. Pearlman says this about montage "Technically, the editing is assemblage of pieces into a whole. Creatively, montage is assembling of images and sounds into relations that generate rhythms, ideas, and experiences of a whole" [14, p.155]. The idea is that the director and the editor can do a lot creatively by assembling clips in different ways, thus varying the tempo amongst other things. Additionally, Pearlman raises the term continuity cutting. The term is described as an event that unfolds in real time and thus could be filmed in a single take but is instead utilizing multiple shots and angles cut together to present the event. Consequently, with multiple shots the editor has the freedom to create various feelings of time and pace.

2.4 Tempo

It is suggested by Dancyger that variations in tempo are generally used for dramatic effect and it directs the viewer's emotional response when watching a film [19]. This is most prominent in action scenes where higher pace creates intensity and vice versa. In his book "*The Technique of Film and Video Editing*" Dancyger [19] describes action in film. The action sequence contains accelerated movement and it is the best way to show how impactful movement can be in film, and this goes for everything from chase scenes to fight scenes. These scenes are elevated by being important points in the film with more speed and dynamics than the average scene. The action sequence should get viewers on the edge and be emotionally involved. Editing should complement emotions such as identification, intensification, and excitation. Excitation is achieved through pans, tilts, zooms and tracking to follow the movement and characters. Additionally, variation in shot lengths or making shots very short will accomplish excitation and intensification.

For the purpose of this study, tempo was established as the speed and the rhythm of movement and cuts. From Pearlman's editing point of view, she excludes the sound in her definition of tempo. Besides overall rate of content, Pearlman describes movement and cuts as the rate of change within a shot and cutting rate respectively [14]. Cutting rate is the rate at which shots occur within a certain time, and the duration of each of these shots. Plenty of cuts within a short period of time gives the sensation of a high tempo. An example of this is if the same scene is cut in different ways. Many cuts will make the scene feel quick and sharp, whereas less cuts leave the shots lingering, for a loose and slower sensation. The other influencing factor on tempo was the rate of change, this refers

to the events and movement that happens within the shot without cutting.

Both cutting rate and movement can be quantified as the two primary contributors to the tempo in a clip or film, as ASL and motion [5], or as ASL and visual activity [15]. To then quantify and measure tempo, both ASL and motion need to be extracted and calculated from a film.

2.5 Tempo formula

In a study by Adams et al. [5], the possibilities to extract expressive elements from film are examined. They provide an effective way to compute the characteristic of tempo in film. The tempo is made into a computable entity by using the function:

$$P(n) = \alpha \frac{(med_s - s[n])}{\sigma_s} + \beta \frac{(m[n] - \mu_m)}{\sigma_m} \quad (1)$$

With the seven parameters in the function being:

s = Shot length in frames.

m = Motion magnitude.

n = Shot number.

σ_s = Standard deviation of shot length.

σ_m = Standard deviation of motion magnitude.

μ_m = Average motion.

med_s = Median shot length.

α and β are weights to make motion and shot length contribute equally to the tempo, depending on different circumstances, such as the genre of the film [5]. In the paper, this formula is used to see how much a movie's tempo deviates from its average throughout the film and when. The plan is to use reference values for the deviation values. This makes it possible to compare how each film deviates from a fictional average rather than itself and makes it possible to compare each film's tempo.

2.6 Movement calculations

Information about the intensity of motion is useful when working with various kinds of video. Different approaches to compute motion are examined by Bhurane et al. One of these is the motion vector-based approach. Though the vector-based approach is computationally heavy, it is a good estimate of the motion. The approach uses the equation:

$$MVM = \frac{1}{N} \sqrt{MV_x^2(k) + MV_y^2(k)} \quad (2)$$

It takes the sum of the length of all vectors and divides by the number of vectors in each frame. This provides the estimated motion in a frame from a video [7].

The extraction of motion vectors is described by Mücklich in his master thesis “*Motion Feature Extraction of Video and Movie Data*” as the following “... motion estimation is carried out by assigning each pixel block of the frame, which is encoded, a motion vector. A motion vector represents a two dimensional vector containing the offset to a reference frame. This offset describes how such a block moves in the two dimensional pixel domain considering a reference and a current frame. This technique is also referred to as inter predictive coding” [21, p.5].

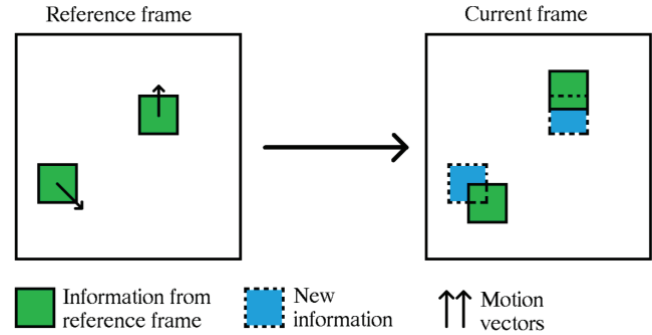


Figure 1: Image showing how motion vectors are calculated

3. METHOD

Here the method used will be presented. First how the particular films were chosen, and how their tempo was calculated with ASL and motion. Second, the test will be described. The participants are presented followed by how the test was conducted. Lastly the questions for the questionnaire are shown.

3.1 Choosing Films

Action movie scenes with high tempo and varying shot length will be shown to our test subjects. All scenes share a high tempo but have different average shot length and movement. When choosing movies for the study, clips had to be from films within the action genre. To make a comparison easier, all the clips contained some sort of chase or running scene. The clips should have either a lot going on in the frame i.e., high motion, or utilize a considerable number of cuts. Most importantly clips with varied shot length should be represented.

3.1.1 Point Break

“*Point Break*” [9] contains a famous chase scene with slightly longer shot length. The scene features a lot of camera movement and camera shake which makes it a good scene to use for the study. In an interview [16], the director Kathryn Bigelow talks about how a lot of the camera work was done handheld and how it created a lot of momentum, working like the human eye by panning and following the action around.



Figure 2: Still images from the films in order from top left, *Point Break*, *Dunkirk*, *Casino Royale*, *Taken 3* and *1917*.

3.1.2 1917

“1917” [10] is a war movie made to seem like it is shot in a single long take. The particular scene chosen from this movie shows the main character running along a battlefield with a lot happening around him. Since the scene has no cuts, it can be classified to the extreme regarding high ASL.

3.1.3 Dunkirk

“Dunkirk” [11] is another war movie that utilizes longer shots. However, it is not shot in a single take but cuts are used sparsely. The chosen clip takes place in the beginning of the film, and the main character is seen running away when being shot at.

3.1.4 Taken 3

“Taken 3” [12] can be seen as the other extreme case. Here the ASL is much shorter than the rest of the clips and more than one cut can be seen during just a second at times.

3.1.5 Casino Royale

From “Casino Royale” [13], the chase scene from the beginning of the film is used. It makes use of wider shots and it is also the only film making more use of vertical movement by characters through the environment. The film has a short ASL, but no extreme motion in the scene.

3.2 Extracting Features

To be able to compare the different scenes in terms of tempo, its ASL and motion had to be calculated. The ASL ensures that the scenes display different types of shot lengths. Motion was measured to give a better idea of whether the scenes have similar tempo despite the shot length, as motion is the other major contributor to the tempo of a scene.

3.3 Motion Calculations

There were different ways to implement a tempo estimation in video, initially the plan was to create a simplistic motion measuring algorithm to extract the motion parameter. The first iteration

consisted of dividing each frame into smaller blocks of pixels, each of these blocks would be given a search window around itself, with a width and height that’s two times as large as the block’s width and height.

The search window was applied in the following frame where it would detect how the block has moved in relation to the previous frame, similarly to figure 1. A problem with the algorithm was that it searched every single block, regardless of if there was any movement which resulted in the algorithm picking up the same block in the following frame at multiple places. For example a frame of the sky will have a lot of blocks that are similar, thus the implementation did not work as intended. Reworking the algorithm to only create a search window for blocks that had moved was a possible solution. In other words the algorithm wouldn’t look at parts of the frame that didn’t change. This could be done by subtracting the block with the block on the same position in the following frame. If there would have been a change in values in between frames, the matrix returned would contain the difference in these values. If all returned values are zeros that means the block is in the same position as in the previous frame.

In order to save time a motion extraction library found on GitHub was used instead of reworking the algorithm. It was created by Bommes at MIT [8], this algorithm had the same idea with extracting motion vectors from the distance a block of pixels have moved but was different in using H.264 compression to extract the motion vectors. This algorithm was used to extract our motion vectors for each shot. With these vectors we could use equation 2 to get a value for the motion in the scene. The value we obtain is called the motion magnitude of the shot and is one of the parameters necessary to calculate the tempo in the scene.

3.4 Extracting ASL

Calculating the ASL of each clip was done by watching them and counting the cuts. The clip length in seconds was then divided by the number of cuts and the ASL was attained.

3.5 Tempo Calculations

The motion magnitude and ASL are then inserted into a modified version of the tempo formula (1), Where $\alpha = -1$, $\beta = -\frac{1}{3}$, $\sigma_s = 1$, $\sigma_m = 1$, $\mu_m = \text{Average motion magnitude}$, $med_s = \log_{10}(ASL)$.

Here, the ASL is in seconds instead of frames and the final version of the formula is as follows:

$$P = \frac{1}{3}\mu_m - \log_{10}(ASL) \quad (3)$$

The average motion and ASL of the entire scene are what is important to extract. Instead of performing the calculation once every shot and looking at how the tempo deviates from its own average and changes throughout the clip, the entirety of the scene is calculated once with the average motion and ASL. Thus, the shot number(n), shot length in frames(s) and the motion magnitude(m) for each shot are not relevant to the study apart from attaining the average motion and ASL.

3.6 Test Structure

The experiment took place both digitally and inside the media production lab at KTH where a projector and sound system had been set up. A total of eleven people were invited to participate in the study (table 1), eight from the media technology program at KTH and three outsiders. Out of these eleven people, four had the opportunity to participate physically, the remaining seven participants got the questionnaire and clips e-mailed to them so they could participate digitally.

Participant	Gender	Age	Place
P1	male	41	physical
P2	male	20	physical
P3	male	20	physical
P4	female	21	physical
P5	female	28	digital
P6	male	20	digital
P7	male	28	digital
P8	male	24	digital
P9	male	26	digital
P10	male	26	digital
P11	male	29	digital

Table 1: List of all the participants gender, age, and if they did the study in person or digitally.

3.6.1 Implementation

Firstly, the participants viewed the questionnaire (see **Appendix A**) they would fill after the viewing. Next a one-minute clip from each of the movies was shown. The film clips were shown in the following order:

1. “*Point Break*”
2. “*1917*”
3. “*Taken 3*”
4. “*Dunkirk*”
5. “*Casino Royale*”.

After all the clips had been viewed the participants filled out the questionnaire which contained information regarding the use of collected data for research and educational purposed, followed by questions of tempo, emotions, preference, et cetera. The questionnaire was written in Swedish, its questions have been translated below.

The first question was

“Have you seen any of the films previously?”

followed by two questions regarding the tempo

“Which of the film sequences do you think had the highest tempo?”

and

“Which of the film sequences do you think had the lowest tempo?”.

The three final questions were:

“Which film do you think had the highest budget?”,

“Did any of the scenes awaken any emotion? For example fear, stress, excitement...”

“Which of the film sequences did you like the most? Motivate your answer”.

We present our qualitative data in forms of quotes that translate general trends in the dataset. Quantitative results are presented as diagrams and percentages.

4. RESULTS

In the first part of this section, the results from the motion and ASL calculations will be presented, followed by the calculated tempo. Then the results from showing the clips to our test subjects will be presented, which shows their expressed feelings and perceived tempo on the films from the questionnaire.

4.1 ASL Results

Film title	ASL
Taken 3	0.580s
Casino Royale	1.40s
Point Break	3.88s
Dunkirk	5.60s
1917	61.0s

Table 2: List showing the films ASL in seconds, from lowest to highest

The results of measuring the motion and the ASL for each film can be seen in table 2. This shows that both “*Taken 3*” and “*1917*” are very far from the rest in terms of shot length. “*Taken 3*” having an ASL of 0.58s means it on average cuts almost twice every second during the sequence. “*1917*” has its full sequence length in time as ASL, because there are no cuts in the scene.

4.2 Motion Magnitude Results



Figure 3: Example of how the motion vectors looks in a frame from *Dunkirk*.

The first part of the results in the motion magnitude calculations came from extracting motion vectors from the clips. Figure 3 shows a visual representation of the motion vectors extracted from one frame. The red arrow (vector) shows where and how far a block of pixels have moved. These vectors are the ones later used in equation 2, where the results provide the average motion for the clip. In figure 4, the films are compared by the average motion. Here, “*Dunkirk*” has the highest motion value compared to the rest of the films, followed by “*1917*”.

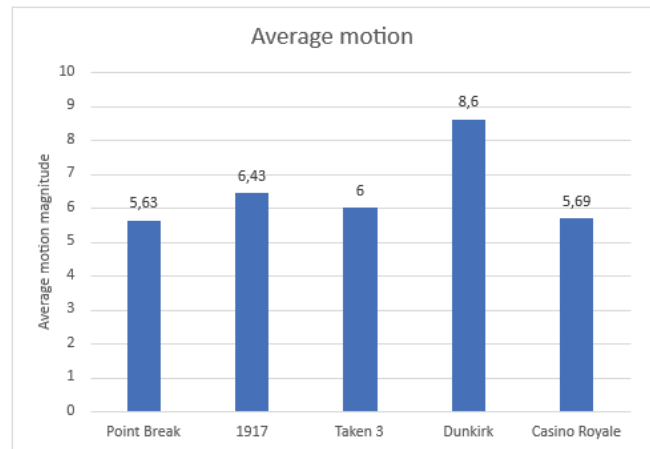


Figure 4: Graph presenting the films and their average motion value

4.3 Tempo results

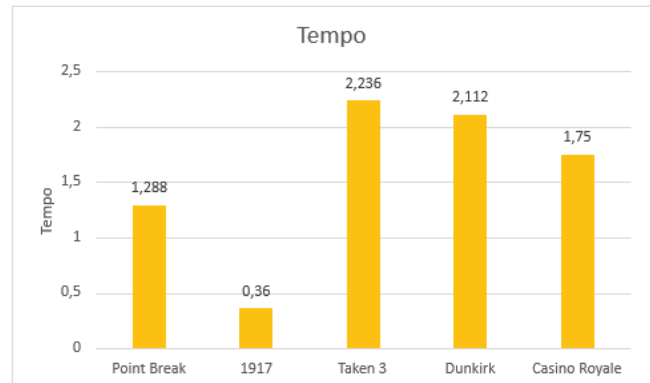


Figure 5: Graph with each films calculated tempo

After motion and shot length were combined in the tempo formula and the results of the tempo calculations were reviewed, some clear results emerged. Firstly, it was clear that “*Taken 3*” had the highest tempo out of the films with a tempo value of 2.236, and secondly that “*1917*” by far had the lowest with a tempo value of 0.36. These results can be seen in figure 5 where each of the film’s final tempo are compared in the graph.

4.4 Relation to the films

From the first question of the questionnaire, which was if the participants had seen any of the films prior, none of the participants had seen all the films before the study. One of the participants (P4) had not seen any of the films, while the rest had seen between one and four of the movies. “*Casino Royale*” was the film most participants had seen prior with six participants saying they had seen it before. The least watched was “*Taken 3*” with two participants saying they had seen it. This question hinted towards a connection between the preference of film and previously watched films. Most participants that preferred “*Casino Royale*” had previously watched it. For “*Taken 3*” no one of the participants preferred it and two had previously watched it. Participant 1 said it

is probably harder to understand the context of the films with low tempo compared to the ones with high tempo if you have not seen it before.

4.5 Perceived Tempo

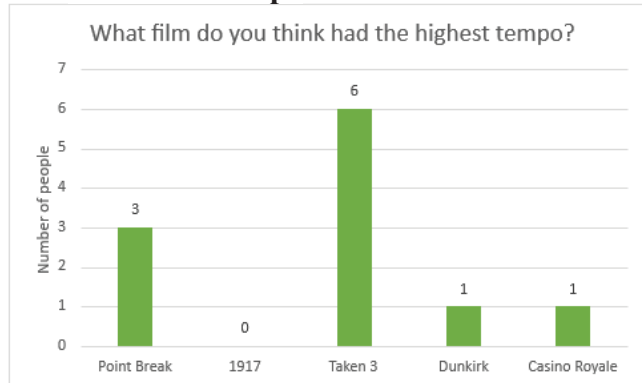


Figure 6: Graph with the number of people who thought each film had the highest tempo.

From how the participants perceived tempo, the results of which film had the highest tempo can be found in figure 6. Here, 6 of the participants (57,1%) thought “Taken 3” had the highest tempo out of the clips. 3 participants (28,6%) thought “Point Break” had the highest tempo. No one of the participants said “1917” had the highest tempo.

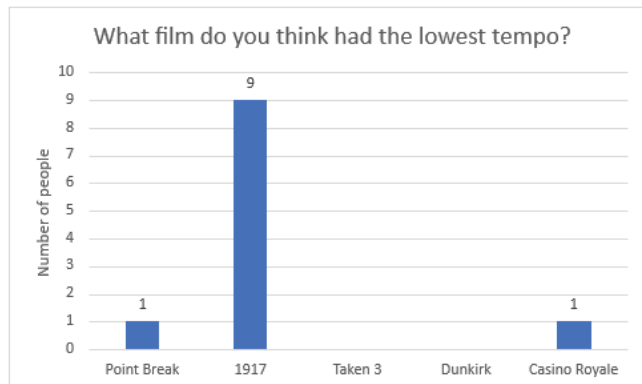


Figure 7: Graph with the number of people who thought each film had the lowest tempo.

Results were more one sided when asked about which clip had the lowest tempo. 71% found “1917” to have the lowest tempo. The two other participants answered “Point Break” and “Casino Royale” respectively.

4.6 Emotions and preference

Participants were asked what emotions they felt watching each sequence and which sequence they preferred. Generally, emotions like excitement, stress and fear were associated with each sequence.

For the sequences with higher ASL (1917, Dunkirk), participants leaned more towards feelings like fear and angst, while on the other hand sequences with low ASL (Taken 3, Casino Royale) participants related more to stress and in some cases excessiveness. For example, participant 5 said “Taken 3 Bored, feels unimaginative and excessive [...] Casino Royale felt nothing special, also a bit excessive”.

For “Point Break” [9], most common feelings were stress and excitement, nevertheless participants did not feel it was as intense as the others yet more realistic.

One participant (P6) thought “1917” [10] was more epic than the rest of the sequences but did not think it was as exciting. Although for the rest of the participants, consensus was the feeling of excitement from the clip. Besides this, other emotions were generally more dramatic.

The common emotion felt from “Taken 3” [12] was stress. About half the participants said stress in general, and the other half said stress in a more negative connotation together with feelings like irritation. Participant 2 deliberated that the stress came from the editing rather than what happened in the clip. “Stress but not from the scene like, more from how it was cut in a way that made it stressful”.

Many of the participants (P3, P4, P5, P6, P11) had some negative unpleasant feelings when watching “Dunkirk” [11]. As an example, the emotions consisted of fear, helplessness, and angst.

Lastly, for “Casino Royale” [13], most of the participants (P2, P3, P4, P6, P8, P10) thought of it as exciting and nothing more. Participants 5 and 11 thought of the scene as more excessive.

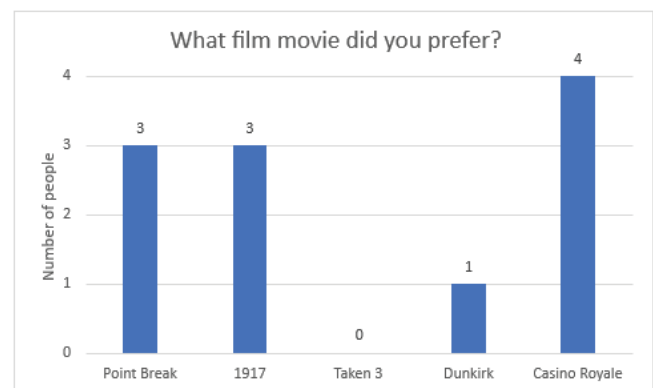


Figure 8: How many participants who preferred each film.

When asked which of the sequences was preferred, 40% answered “Casino Royale” [13]. The participants reasoning as to why were: “The camera was steady and it was easy to follow but still had a lot of movement and energy” (P1), “I liked the change of scenery and

that it contained less jump cuts” (P4),” Cool scene with impressive parkour” (P10)

Out of all participants 30% preferred “*Point Break*” [9] with the reasoning being : “I enjoyed the film techniques and the scene had character” (P5), “It felt more real, the stunts were not as sick but that didn’t make the scene less exciting” (P6), “Easy to understand what is going on and the tempo was high” (P9).

The final 30% of participants preferred “*1917*” [10] with the reasoning being: “It was the most powerful and because it’s shot like a long sequence it felt more realistic” (P11), “I haven’t seen this movie before, but I thought it was the most exciting. I wonder what he was running towards in the end” (P4), “I preferred *1917* probably because I watched a video that explained behind the scenes” (P2).

In the last section of the questionnaire participants were asked which of the films they thought had the largest budget. 55% of the participants thought that “*1917*” [10] had the largest budget. The remaining 45% thought that “*Casino Royale*” [13] had the largest budget.

5. DISCUSSION

5.1 The final formula

In the original tempo formula (Equation 1) from Adams et al. [5] the shot length is measured in frames instead of seconds, thus when using the same formula with seconds the result of the shot length will be 24 times smaller. This is because the framerate of all the clips was 24 frames per second. Therefore, the ASL would impact the pace much more than the motion. With our approach we scale down the average motion to a point where the $\log_{10}(ASL)$ impacts the tempo in a way that suits our study, the suitable weight for β was $-\frac{1}{3}$, that way the average motion is set to a positive and scaled down. The ASL goes through a logarithmic transform, this is suitable because deviations in low ASL have a much bigger impact than a deviation from a high ASL. Also, ASL under one second should contribute to a higher tempo while higher ASL should reduce the tempo. The final result was equation 3.

5.2 Problems

5.2.1 Impact of Sound

In this study the impact of the sound was neglected, but its impact on the results cannot be ignored. All clips featured had background music to enhance the experience, the most subtle is *Dunkirk* which has a Shepard's tone [22] playing throughout it to enhance the tension. The Shepard's tone is significantly lower than the background music in the others clips but may still have affected the results, especially for the participants who performed the test digitally as we cannot confirm what type of sound system they used or at what volume. For example participant 8 described *Dunkirk* as less stressful because it lacked background noise.

5.2.2 Problems with digital participation

There were no distinguishable differences in the results of the questionnaire when comparing physical and digital participants, but it’s impossible for us to know for sure that everyone that participated digitally did so in a desirable fashion. A possibility is that some or all digital participants didn’t watch any of the clips and just filled out the questionnaire. It would have been beneficial to have additional questions for digital participants such as “what sound system did you use when watching the clips?” and “What size and resolution is the screen you used to watch the clips?”. This would have given more insight into what environment the participants watched the clips and if it had any impact on their immersion and emotional response.

5.2.3 Problems with the motion method

The prediction for the motion vectors was that “*Dunkirk*”, “*1917*” and “*Point Break*” would score higher, this is because they all had higher ASL and thus needed to have more motion to create a faster pace. From the results we can see that this was not the case for “*Point Break*” and there are possible reasons for this, one being the camera quality. “*Point Break*” is older than the rest of the films included, it was made in 1991 and the second oldest movie was “*Casino Royale*” from 2006. Due to being older the camera quality is not as good as in the newer movies, also the scene from “*Point Break*” was shot with a handheld camera with a lot of 180 degree turns, with a frame rate of 24 frames per second that results in some blurry frames (see figure 9).



Figure 9: Image of blurry frames from *Point Break*

5.2.3.1 Alternative motion method

Because of this the algorithm could have had difficulties detecting some of the motion in “*Point Break*”. A better alternative for extracting the motion from “*Point Break*” would have been to use the visual activity index specified by Dokic et al. [15]. In the method frames are scaled down and turned into grayscale, giving each pixel a value from 0 to 255. Changes in pixel values are then examined from one frame to another. This method could have been used for all the clips, that way the motion magnitude would be limited to a value between zero and one, thus giving a better

representing value for all the films and there would be no need to scale down the β -weight in equation 3, at least not as much.

5.3 Discussing tempo

5.3.1 Calculated tempo

The results from the ASL (Table 2) show “*Taken 3*” and “*1917*” being very far from the median. This makes the other three films seem like they have a more similar ASL. However, this is not really the case as they are all very distinct in shot length. It was expected that “*Taken 3*” would score the highest when the extracted parameters were inserted into the tempo formula (Equation 3) and that “*1917*” would score the lowest because those were the extreme cases of ASL, however there was no prediction of where in the middle “*Dunkirk*”, “*Point Break*” and “*Casino Royale*” would end up. Even though the score of “*1917*” was expected to be low, it wasn’t necessarily expected to score as low as it did, this is in large because its ASL was drastically higher than the rest of the clips.

5.3.2 Participants perceived tempo

From the question regarding the perceived tempo of the film, it is obvious that the ASL played a big part in it. In the question about highest tempo, “*Taken 3*” with its extremely low ASL of 0.58s had most participants thinking it had the highest tempo. The same goes for lowest tempo where mostly everyone thought “*1917*” with no cuts at all had the lowest tempo. Comparing this to the calculated tempo, both “*Taken 3*” and “*1917*” had the highest respectively the lowest values in our calculations. This means the calculations coincided with the perceived tempo from the participants. However, when comparing the participants perceived tempo to the calculated motion, “*Taken 3*” had the third highest motion and “*1917*” had slightly higher. Since the motion values between these two were very similar, it makes sense it would not have a big impact when comparing the two films’ tempo.

The very fast cutting in “*Taken 3*” was obviously something the participants noticed when watching the clip, and mostly in a negative light thinking it was too hectic. In the same way, the dolly shot in “*1917*” only showed one angle of a character running without any change. So even though a lot happened with lots of motion in the frame around the main character, it is still perceived as very slow because of its high ASL.

The other three films were not as frequently mentioned by participants regarding tempo. “*Point Break*”, “*Casino Royale*”, and “*Dunkirk*” were all answered once in both questions about which had the highest and lowest tempo. The only exception here being “*Point Break*” which had three answers for it having the fastest tempo. None of these three movies were extreme in any way, yet still having distinct ASL and camera movement. Despite this, there was no way to distinguish them by the results of the questionnaire, as “*Taken 3*” and “*1917*” were the inevitable answers. All things considered, a better way to conduct the experiment would have been to not use the outliers regarding ASL. Five clips were a bit too many to compare, and only using the three clips would result in fairer comparison and more appropriate results. It would also put more emphasis on the motion in conjunction with ASL in viewing experience, without the dramatically apparent differences from the outliers. Another idea for the tempo question would have been to

rank all clips’ tempo instead. This way there would have been a lot more information to gather from all the clips.

5.3.3 Participants preferences and emotions

The questions about preference and emotions provided some useful insight on how tempo affects how viewers experience a film sequence. A general theme in emotions for the films with high ASL were more dramatic emotions and not as intense. These films were “*1917*” and “*Dunkirk*”, both are war movies. The war theme is a possible reason for why viewers felt for example fear when watching the clips. It might also be the other way around, that the specific type of editing style was used to evoke these types of feelings when watching the films. For example, the feeling of identification [19] seems to be more present with these scenes, which could be due to the more lingering shots [14] and the camera work being more personal.

It could be inferred that the threshold for low ASL is somewhere between “*Taken 3*” (0.58s) and “*Casino Royale*” (1.4s). Hence feelings being more negative towards “*Taken 3*” and more people preferring “*Casino Royale*”. Going past 1.4 seconds would mean it could be perceived as irritating or annoying from what the participants answered. This also relates to the question about what movie had the highest budget. Out of the participants, 45% replied that they thought “*Casino Royale*” had the highest budget of the films, with the other 55% saying “*1917*” had the highest budget. A question that in hindsight should have been asked is what film the participants thought had the lowest budget, this would have provided more insight into the link between resources and tempo.

5.3.4 Context and tempo

Something that became clear when reading the results was the connection between tempo and context. All clips used were around one minute long and didn’t provide any information on the story. The story itself can be a tempo raising factor, an example of this can be a still shot of people sitting around a table. That would seem rather innocent to someone without knowledge of the story, but if say a bomb was hidden under the table and the viewer knows this, the scene would feel more intense. Therefore, scenes with a lot of tempo raising editing or filming techniques are not in need of context to have a high tempo. Scenes without a lot of tempo raising elements on the other hand relies more on its context to raise the tempo and create intensity in the scene.

6. CONCLUSION

From the two prominent features explored in this study, motion did not have much impact compared to the ASL. Trends in tempo and emotion from the participants could be seen as connected to mainly ASL, with high ASL resulting in low perceived tempo and certain emotions. From this it can be suggested that when editing a film with continuity cutting and having the shot lengths shorter, the viewer will perceive the film as having higher tempo, as mentioned by multiple sources [5][14][15][19]. Furthermore, it is important to take into account that a higher tempo through a low ASL will not always be favored if the ASL is too low.

With both motion and ASL contributing to tempo, either or both our calculations and tests could be critiqued. The weights α and β could have been further adjusted to suit the study, this would have taken more time to fix but could have resulted in more accurate tempo calculations, especially for the three films that was in the middle regarding tempo. A problem was that the films with the highest and lowest tempo from our calculations, were also the films with the highest and lowest ASL respectively. This means we cannot fully conclude if ASL is the main impact on experienced tempo due to the formation of our test. However, from our test it is suggested ASL definitely has an impact on the viewing experience.

We can't conclude anything from the study, the dataset was far too small to make any generalizations, however the study provides a good base for further study on the subject and hints towards patterns that can be interesting for future papers on cinematic tempo.

7. REFERENCES

- [1] Murphy, Arthur D. , Andrew, Dudley , Manvell, Roger , Sklar, Robert and Stephenson, Ralph. "film". Encyclopedia Britannica, 3 Feb. 2022, <https://www.britannica.com/art/motion-picture>. Accessed 3 March 2022.
- [2] Nash information services. The numbers. 2022 <https://www.the-numbers.com/market/genres> Accessed 5 March 2022
- [3] Bordwell, David. 2006. The way Hollywood tells it. Berkeley, CA: University of California Press.
- [4] Cutting, James & Candan Simsek, Ayse. 2015. Shot Durations, Shot Classes, and the Increased Pace of Popular Movies. Projections..
- [5] Adams, B., Dorai, C., & Venkatesh, S. (2002). Toward automatic extraction of expressive elements from motion pictures: Tempo. IEEE Transactions on Multimedia, 4(4), 472-481.
- [6] M. V. Srinivasan, S. Venkatesh, and R. Hosie, "Qualitative extraction of camera parameters," Pattern Recognit., vol. 30, no. 4, pp. 593–606, 1997
- [7] Bhurane, A. A., Nishar, H., Agarwal, H., Kothari, A., & Gadre, V. M. 2016, March. Spectral approaches for motion intensity estimation in video coding. In 2016 Twenty Second National Conference on Communication (NCC) (pp. 1-6). IEEE.
- [8] Bommès, L., Li, X., Zhou, J. 2020. 15th IEEE Conference on Industrial Electronics and Applications (ICIEA), MVmed: Fast Multi-Object Tracking in the Compressed Domain. p(1419-1424), 10.1109/ICIEA48937.2020.9248145
- [9] Bigelow, K. 1991. Point Break. USA: Largo Entertainment.
- [10] Mendes, S. 2019. 1917. USA: DreamWorks Pictures.
- [11] Nolan, C. 2017. Dunkirk. UK: Syncopy.
- [12] Megaton, O. 2014. Taken 3. France: EuropaCorp.
- [13] Campbell, M. 2006. Casino Royale. USA: Columbia Pictures.
- [14] Pearlman, K. Cutting Rhythms, 2nd Edition. 2016.
- [15] Dokic, K., Mandusic, D., Blaskovic, L. (2020). Analysis of Hollywood's Film Production Using Background Subtraction Visual Activity Index Based on Computer Vision Algorithm. In: , et al. New Trends in Information and Communications Technology Applications. NTICT 2020. Communications in Computer and Information Science, vol 1183. Springer, Cham. https://doi.org/10.1007/978-3-030-55340-1_17
- [16] VHS Video vault. 2019, July 5. Moving Pictures - Kathryn Bigelow on Point Break [Video]. YouTube. https://www.youtube.com/watch?v=jXm1iDrofEM&ab_channel=VHSAudiovault
- [17] Britannica, T. Editors of Encyclopedia (2016, March 21). montage. Encyclopedia Britannica. <https://www.britannica.com/art/montage-filmmaking>
- [18] Nothelfer, Christine E., Jordan E. DeLong, and James E. Cutting. "Shot Structure in Hollywood Film." (2009).
- [19] Dancyger K. T. The Technique of Film and Video Editing. 2013.
- [20] Gannon, E. T., & Grubb, M. A. (2022). How filmmakers guide the eye: The effect of average shot length on intersubject attentional synchrony. Psychology of Aesthetics, Creativity, and the Arts, 16(1), 125–134. <https://doi.org/10.1037/aca0000315>
- [21] Müglich, M. (2017). Motion Feature Extraction of Video and Movie Data (Dissertation). Retrieved from <http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-214030>
- [22] Shepard, Roger N. (December 1964). "Circularity in Judgements of Relative Pitch"

Appendix A – Questionnaire

KexStudie 22 April

Logga in på Google för att spara förloppet. [Läs mer](#)

***Obligatorisk**

Har du sett någon av filmerna tidigare? *

- ☐ Film 1 - Point Break
- ☐ Film 2 - 1917
- ☐ Film 3 - Taken 3
- ☐ Film 4 - Dunkirk
- ☐ Film 5 - Casino Royal
- ☐ Har inte sett någon av ovanstående filmer

Vilken av filmsekvenserna uppfattade du hade HÖGST tempo? *

- ☐ Film 1 - Point Break
- ☐ Film 2 - 1917
- ☐ Film 3 - Taken 3
- ☐ Film 4 - Dunkirk
- ☐ Film 5 - Casino Royale

Vilken av filmsekvenserna uppfattade du hade LÄGST tempo? *

- ☐ Film 1 - Point Break
- ☐ Film 2 - 1917
- ☐ Film 3 - Taken 3
- ☐ Film 4 - Dunkirk
- ☐ Film 5 - Casino Royal

Vilken av filmerna tror du hade störst budget? *

- ☐ Film 1 - Point Break
- ☐ Film 2 - 1917
- ☐ Film 3 - Taken 3
- ☐ Film 4 - Dunkirk
- ☐ Film 5 - Casino Royale

Väckte någon scen några känslor hos dig? Exempelvis rädsla, stress, spänning.... *
(Skriv gärna något om varje film)

Ditt svar

Vilken av filmsekvenserna föredrog/tyckte du bäst om? Motivera kort varför. *

Ditt svar

Jag godkänner att denna information används i forskning- och utbildningssyften *

☐ Jag godkänner

Din E-mail (kth adress) *

Ditt svar

Skicka

Rensa formuläret

